Low-Complexity Near-Field Localization with XL-MIMO Sectored Uniform Circular Arrays

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Slides & Codes: https://github.com/scliubit/XL-MIMO-sUCA-Loc

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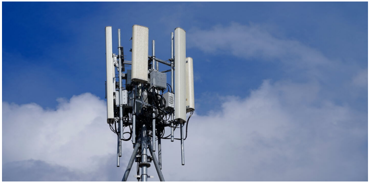
Outlines

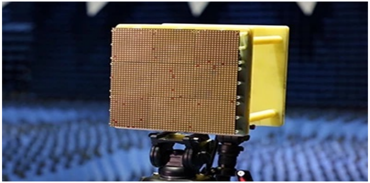
Outline

Introduction

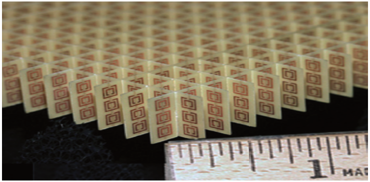
- XL-MIMO antenna arrays have been widely investigated in recent years



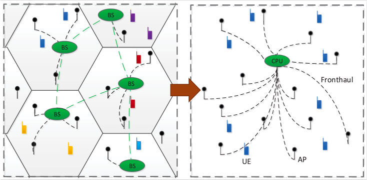
(a) Massive MIMO BS



(b) Recofigurable Intelligent Surface



(c) Holographic MIMO



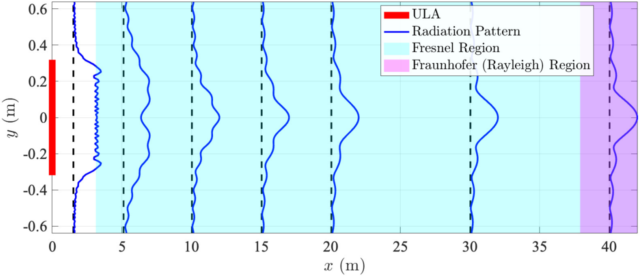
(d) Cell-free mMIMO

Figure 1: Widely investigated XL-MIMO related topics.

Introduction

- Proliferation of antennas (in XL-MIMO) results in increased **aperture**, together with smaller **wavelength** extends the **near-field region** [1]

$$\sqrt[3]{\frac{2D^4}{\boldsymbol{\lambda}}} \leq d_{\mathrm{NF}} \leq \frac{2D^2}{\boldsymbol{\lambda}}, \qquad D\uparrow \ \boldsymbol{\lambda}\downarrow \Rightarrow d_{\mathrm{NF}}\uparrow$$



- Distance-dependent beam pattern poses challenges for wireless systems in the near field

[1] J. Goodman, *Introduction to Fourier Optics* (McGraw-Hill physical and quantum electronics series). W. H. Freeman, 2005, ISBN: 9780974707723.

Introduction

- In the **near-field** region, EM wavefront is not planar, but **spherical**
- **Distance** r and **Angle** θ determine near-field channel together

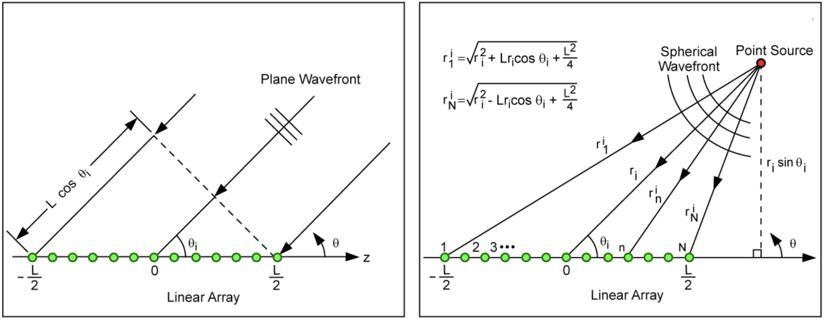
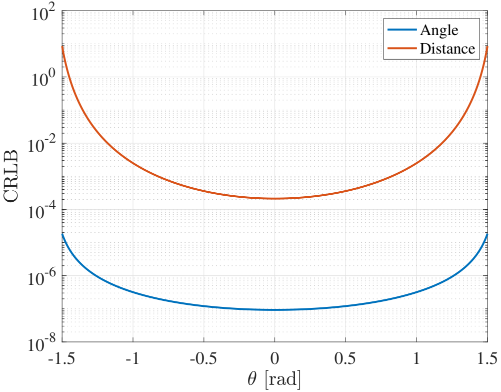


Figure 2: Wavefront illustration [2].

[2] A. Fenn, "Evaluation of adaptive phased array antenna, far-field nulling performance in the near-field region," *IEEE Trans. Antennas Propag.*, vol. 38, no. 2, pp. 173–185, 1990.

Introduction

- Linear (Planar) Arrays are not good at sensing
  - Resolution drains at **edges**.
  - **Range** sensing performance worse than angle's.



Previous Works

Emerging works for utilizing circular arrays:

- Uniform Circular Array (UCA) for azimuthally consistent resolution [3].
- Utilizing UCA for reduced codebook size [4].
- Rotational symmetry of UCA provides uniform and enlarged near-field regions at all angle [5].
- Near-field beamspace beamforming with UCA [6].

[3] T. Liang, M. Zhu, and F. Pan, "A doa estimation method for uniform circular array based on virtual interpolation and subarray rotation," *IEEE Access*, vol. 9, pp. 116 760–116 767, 2021.

[4] Y. Xie, B. Ning, *et al.*, "Near-field beam training in thz communications: The merits of uniform circular array," *IEEE Wireless Communications Letters*, vol. 12, no. 4, pp. 575–579, 2023.

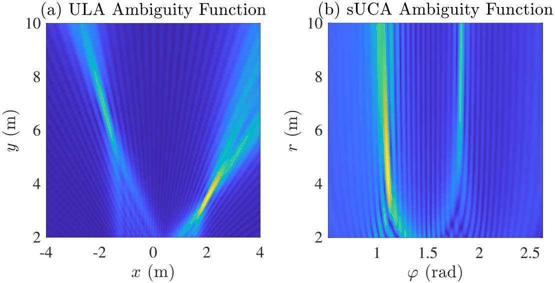
[5] Z. Wu, M. Cui, and L. Dai, "Enabling more users to benefit from near-field communications: From linear to circular array," *IEEE Transactions on Wireless Communications*, vol. 23, no. 4, pp. 3735–3748, 2024.

[6] F. Zhang and W. Fan, "Near-field ultra-wideband mmwave channel characterization using successive cancellation beamspace uca algorithm," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 8, pp. 7248–7259, 2019.

Previous Works

Our motivations:

- Better beam pattern representation for localization



- Consistent resolution with improvements in distance
- Fast positioning with FFT

Outline

System Model

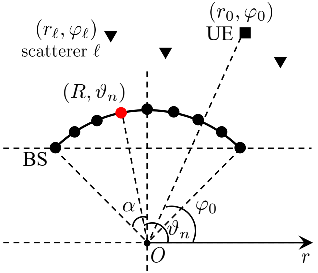


Figure 3: Sectored uniform circular array (sUCA) with N antennas.

- Near-field received signal at the k-th subcarrier f_k

$$\mathbf{h}_k[n] = \frac{1}{\sqrt{NL}} \sum_{\ell=0}^{L} \frac{\alpha_\ell e^{\jmath \frac{2\pi}{c} f_k (d_\ell + d_{\ell,n})}}{d_\ell d_{\ell,n}} = \sum_{\ell=0}^{L} \frac{\tilde{\alpha}_{k,\ell} e^{\jmath \frac{2\pi}{c} f_k d_{\ell,n}}}{d_{\ell,n}}, \tag{1}$$

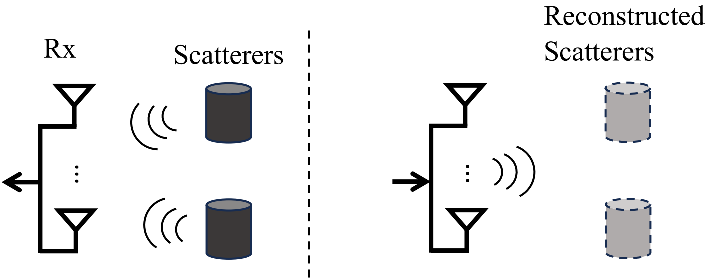
where

$$d_{\ell,n} = \sqrt{r_\ell^2 + R^2 - 2Rr_\ell \cos(\vartheta_n - \varphi_\ell)} \approx r_\ell + \frac{R^2}{2r_\ell} - R\cos(\vartheta_n - \varphi_\ell) \tag{2}$$

Outline

Backprojection

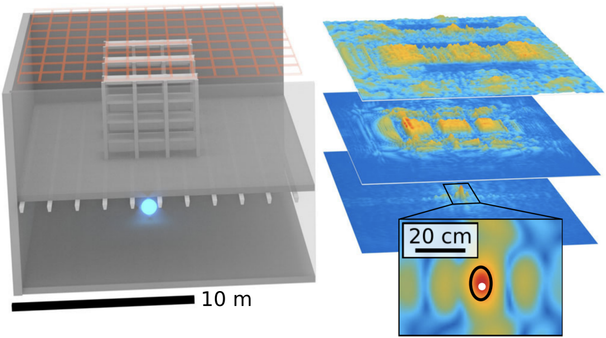
- Backprojection algorithm reconstruction the target by emulating the back propagation of the EM wave



(a) Recording Wavefront (b) **Emulate** Wavefront Back-projection

Backprojection

- It has been widely adopted in sensing/imaging applications [7]



[7] P. M. Holl and F. Reinhard, "Holography of wi-fi radiation," *Phys. Rev. Lett.*, vol. 118, p. 183 901, 18 May 2017.

Backprojection Reconstruction

- Emulated backprojection signal, a.k.a., the *ambiguity function*

$$\begin{aligned} F_k(r,\varphi) &= \sum_{n=1}^{N} \underbrace{\mathbf{y}_k[n] s_k^*}_{\text{Wavefront}} \ \underbrace{e^{-\jmath \frac{2\pi}{c} f_k d_n}}_{\text{Propag.}} = \sum_{\ell=0}^{L} \sum_{n=1}^{N} \frac{\tilde{\alpha}_{k,\ell} e^{\jmath \frac{2\pi}{c} f_k (d_{\ell,n} - d_n)}}{d_{\ell,n}} + \tilde{\mathbf{n}}_k \\ &\triangleq \sum_{\ell=0}^{L} F_k^{(\ell)}(r,\varphi) + \tilde{\mathbf{n}}_k \end{aligned} \tag{3}$$

  at arbitrary coordinate (r,φ) and the k-th subcarrier $f_k = f_c + (k/K) f_{\mathrm{SCS}}$.
- Usually, subcarrier spacing $f_{\mathrm{SCS}} = 15\,\mathrm{kHz}$

- We proved two basic propositions regarding $F_k^{(\ell)}(r,\varphi)$ in (3)

Propositions

Proposition 1

There exists only one unique peak for each $F_k^{(\ell)}(r,\varphi)$ at corresponding (r_ℓ,φ_ℓ)

$$\left|F_k^{(\ell)}(r,\varphi)\right|^2 \overset{(a)}{\leq} \sum_{n=1}^{N} \frac{\left|e^{\jmath\frac{4\pi}{c}f_k R\cos(\vartheta_n-\varphi_\ell)}\right|^2}{\hat{d}_{\ell,n}} \sum_{n=1}^{N} \frac{\left|e^{-\jmath\frac{4\pi}{c}f_k R\cos(\vartheta_n-\varphi)}\right|^2}{\hat{d}_{\ell,n}} \tag{4}$$

Proposition 2

For points $(r,\varphi)\neq(r_\ell,\varphi_\ell)$, $F_k^{(\ell)}(r,\varphi)$ vanishes asymptotically as $f_k\to\infty$ and $N\to\infty$

Demonstration

- Demonstration of the two propositions.

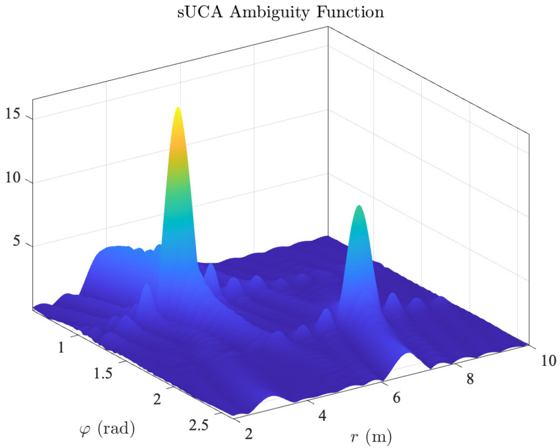


Figure 4: Two targets located at $(4, \frac{\pi}{3})$ and $(7, \frac{7\pi}{12})$, respectively.

FFT Implementation

- Notice that for given distance r we can rewrite (3) as

$$F_k\left(\varphi \mid r, r_\ell, \varphi_\ell\right) = \sum_{n=1}^{N} \mathbf{y}_k[n] e^{-\jmath \frac{2\pi}{c} f_k \sqrt{r^2 + R^2 - 2Rr\cos(\vartheta_n - \varphi)}}. \tag{5}$$

- Define

$$\overline{\mathbf{e}}_{r^{(j)}} = \left[e^{\jmath \frac{2\pi f_k}{c} \sqrt{\tilde{r} - 2Rr\cos\left(\varphi^{(1)}\right)}}, \cdots, e^{\jmath \frac{2\pi f_k}{c} \sqrt{\tilde{r} - 2Rr\cos\left(\varphi^{(G_a)}\right)}} \right], \tag{6}$$

the near-field reconstruction can be implemented by FFT as

$$\mathbf{\Theta}_k[:, j] = \mathcal{F}_{G'_a}^{-1}\left[\mathcal{F}_{G'_a}\left[\overline{\mathbf{y}}_k\right] \odot \mathcal{F}_{G'_a}\left[\overline{\mathbf{e}}_{r^{(j)}}\right]\right]_{N:G'_a - N + 1}, \tag{7}$$

Performance

In short, with reasonable approximations,

- Angular resolution (mainlobe width)

$$\Delta\varphi = \frac{\lambda}{2R\sin\alpha} = \frac{\lambda}{2R} \cdot \frac{1}{\sin\alpha}, \tag{8}$$

whereas ULA:

$$\Delta\varphi = \frac{2\lambda}{\pi R\cos\theta} = \frac{\lambda}{2R} \cdot \frac{4}{\pi\cos\theta} \tag{9}$$

- Distance resolution (mainlobe width)

$$\Delta r = \frac{2c}{f_{\mathrm{BW}}}, \tag{10}$$

whereas ULA beampattern does not show zero-crossing points in the near-field.

Algorithm

Require: $\{\mathbf{y}_k\}_{k=1}^{K}$, grids G_a and G_d, α, distance grid range $[r_{\min}, r_{\max}]$.
Ensure: The estimated coordinates of the UE and scatterers.

for $k = 1, \cdots, K$ **do**
  Initialize empty matrix $\boldsymbol{\Theta}_k \in \mathbb{R}^{G_a \times G_d}$.
  for $j = 1, \cdots, G_d$ **do**
    Fill the j-th column of $\boldsymbol{\Theta}_k$ according to (7).
  end for
  Normalize $\boldsymbol{\Theta}_k$ with

$$\boldsymbol{\Theta}_k := \boldsymbol{\Theta}_k / \max\left(|\boldsymbol{\Theta}_k|\right).$$

end for
Find top $L+1$ indices of peaks $\{i_\ell\}_{\ell=0}^{L}$ from

$$\boldsymbol{\Theta}_a = \sum_{k=1}^{K} \sum_{j=1}^{G_d} |\boldsymbol{\Theta}_k[:, k]|.$$

Find peak on distance grids from

$$j_\ell = \arg\max_j \left| \sum_{k=1}^{K} \boldsymbol{\Theta}_k[i_\ell, j] \right|.$$

Angle estimations $\hat{\varphi}_\ell = \pi/2 - \alpha + 2\alpha i_\ell / G_a$.
Distance estimations $\hat{r}_\ell = r_{\min} + (r_{\max} - r_{\min}) j_\ell / G_d$.

Outline

Simulation Setup

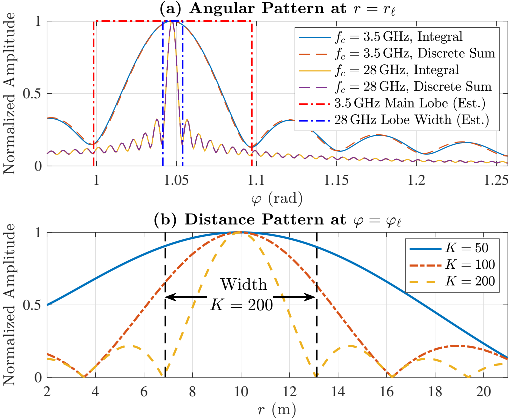
- Evaluation criteria

$$r_{\mathrm{err}} = \frac{1}{L+1}\sum_{\ell=0}^{L}\sqrt{\hat{r}_\ell^2 + r_\ell^2 - 2\hat{r}_\ell r_\ell \cos\left(\hat{\varphi}_\ell - \varphi_\ell\right)},$$

- Sector size $2\alpha = \frac{2\pi}{3}$, near field range $[2\ \mathrm{m}, 21\ \mathrm{m}]$
- $f_c = \{3.5, 10, 28\}\ \mathrm{GHz}$
- Number of subcarriers $K = \{50, 100, 200\}$

Numerical Results

- Reconstruction pattern

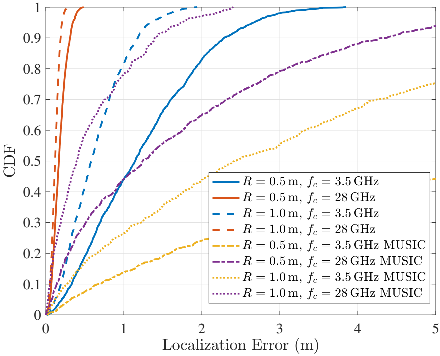


Simulation and Theoretical Results are Consistent

Numerical Results

MUSIC performance is affected due to

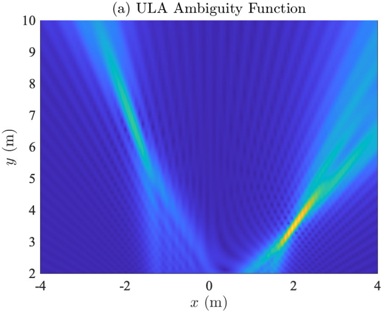
- Steering matrix of sUCA no longer satisfies Vandermonde form
- Two parameters mutually coupled in the steering vector
- Original MUSIC is not an asymptotically consistent estimator for large N



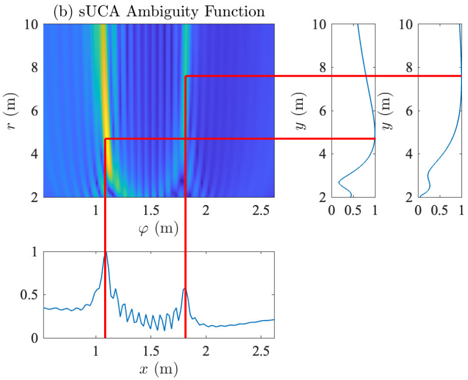
Outperforms MUSIC in sUCA

Example

- Localization can now be achieved by successive 1-D grid search with lower complexity



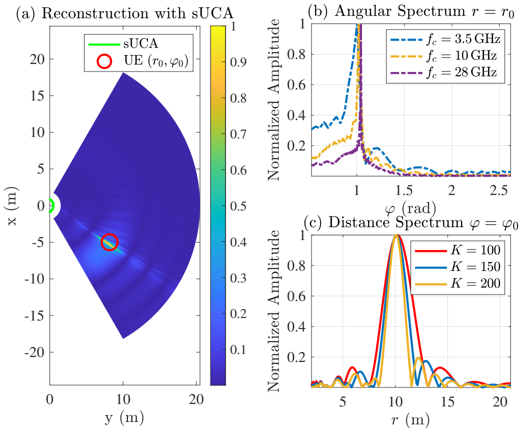
(a) ULA Beampattern



(b) sUCA Beampattern

Numerical Results

- Demonstration on near-field reconstruction



Numerical Results

- Complexity analysis

	Number of Multiplications	Avg. Runtime (s)	
		3.5 GHz	28 GHz
MUSIC	$KN^2 + N^3 + G_a G_d \left((N - L + 1)N^2 + N\right)$	0.5071	58.8952
Proposed	$KG_d(G_a + 1)N$	0.5963	0.7995
Proposed (FFT)	$KG_d\lceil\log_2 G_a\rceil 2^{\lceil\log_2 G_a\rceil}$	**0.0787**	**0.5879**

Proposed Method Shows Lowest Complexity

Summary

- Resorting to BP algorithm for near-field localization
  - Arrays with **Linear** geometry are not suitable for sensing, for its performance drain at edges and worse distance resolution.
  - **Circular** array, as an alternative for ULA, performs better with constant resolution and lower algorithm complexity.
  - Simulations validate and demonstrate our perspectives in near-field sensing.

Outline

[1] J. Goodman, *Introduction to Fourier Optics* (McGraw-Hill physical and quantum electronics series). W. H. Freeman, 2005, ISBN: 9780974707723.

[2] A. Fenn, "Evaluation of adaptive phased array antenna, far-field nulling performance in the near-field region," *IEEE Trans. Antennas Propag.*, vol. 38, no. 2, pp. 173–185, 1990.

[3] H. Wang, Z. Xiao, and Y. Zeng, "Cramér-rao bounds for near-field sensing with extremely large-scale MIMO," *IEEE Transactions on Signal Processing*, vol. 72, pp. 701–717, 2024.

[4] T. Liang, M. Zhu, and F. Pan, "A doa estimation method for uniform circular array based on virtual interpolation and subarray rotation," *IEEE Access*, vol. 9, pp. 116 760–116 767, 2021.

[5] Y. Xie, B. Ning, L. Li, and Z. Chen, "Near-field beam training in thz communications: The merits of uniform circular array," *IEEE Wireless Communications Letters*, vol. 12, no. 4, pp. 575–579, 2023.

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[8] P. M. Holl and F. Reinhard, "Holography of wi-fi radiation," *Phys. Rev. Lett.*, vol. 118, p. 183 901, 18 May 2017.

[9] M. Desai and W. Jenkins, "Convolution backprojection image reconstruction for spotlight mode synthetic aperture radar," *IEEE Trans. Image Process.*, vol. 1, no. 4, pp. 505–517, 1992.

[10] S. Li, S. Wang, M. G. Amin, and G. Zhao, "Efficient near-field imaging using cylindrical mimo arrays," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 6, pp. 3648–3660, 2021.

[11] D. Gabor, "Microscopy by Reconstructed Wave-Fronts," *Proceedings of the Royal Society of London Series A*, vol. 197, no. 1051, pp. 454–487, 1949.

[12] S. Liu, X. Yu, Z. Gao, J. Xu, D. W. K. Ng, and S. Cui, "Sensing-enhanced channel estimation for near-field XL-MIMO systems," *IEEE J. Sel. Areas Commun.*, 2024, *to appear*.

Thanks for your attention!
Q & A

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Backprojection

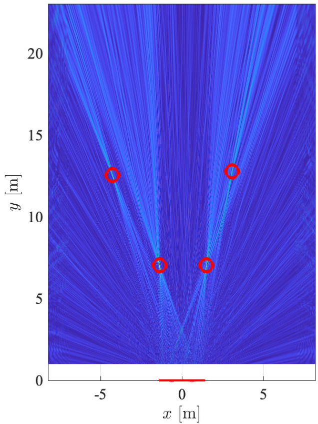
- Backprojection (BP) algorithm has been widely adopted in image reconstruction tasks [8], [9].
- It propagates the received wave front in space according to the EM wave diffraction or angular-spectrum relation [10].
- Example: Reconstruction from z_0 plane:

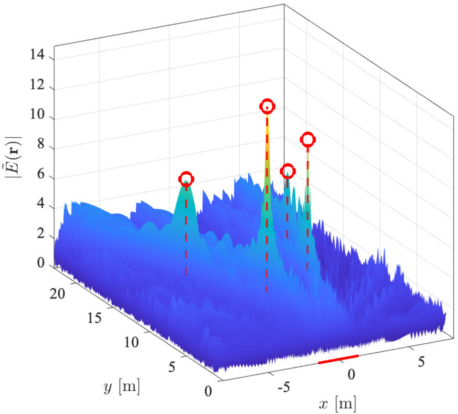
$$I(x,y,z) = \mathcal{F}^{-1}\left[\underbrace{e^{\jmath\frac{2\pi(z-z_0)}{\lambda}\sqrt{1-\lambda^2 f_x^2-\lambda^2 f_y^2}}}_{\mathcal{F}[\text{EM Prop Kernel}]} \times \underbrace{\mathcal{F}\left[I_0\left(x,y,z_0\right)\right]}_{\text{Wavefront}}\right] \quad (11)$$

[8] M. Desai and W. Jenkins, "Convolution backprojection image reconstruction for spotlight mode synthetic aperture radar," *IEEE Trans. Image Process.*, vol. 1, no. 4, pp. 505–517, 1992.
[9] S. Li, S. Wang, *et al.*, "Efficient near-field imaging using cylindrical mimo arrays," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 6, pp. 3648–3660, 2021.
[10] D. Gabor, "Microscopy by Reconstructed Wave-Fronts," *Proceedings of the Royal Society of London Series A*, vol. 197, no. 1051, pp. 454–487, 1949.

Backprojection

- Reconstruction example [11]





[11] S. Liu, X. Yu, *et al.*, "Sensing-enhanced channel estimation for near-field XL-MIMO systems," *IEEE J. Sel. Areas Commun.*, 2024, *to appear*.

Reference Wave

- Obtain precise coherent reference wave (carrier) by phase lock loop (PLL)

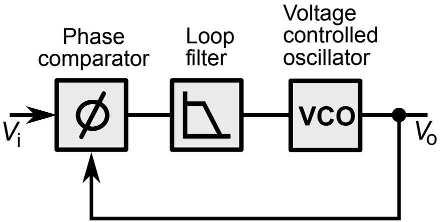


Figure 7: Simple analog phase locked loop